METRIC

DOD-D-23443B
AMENDMENT 2
27 October 1995
SUPERSEDING
AMENDMENT 1
23 September 1985

MILITARY SPECIFICATION

DI-(2-ETHYLHEXYL) ADIPATE, TECHNICAL (METRIC)

This Amendment forms a part of DOD-D-23443B dated 15 May 1978, and is approved for use by all Departments and Agencies of the Department of Defense.

PAGE 1

* Delete: "MIL-STD-1218 ACS Chemicals"

* DD Form 1426 note: Delete "Commanding Officer, Naval Ordnance Station, Standardization Division (611). Indian Head, MD 20640" and substitute "Commander, Indian Head Division, Naval Surface Warfare Center, Standardization Branch (Code 8420), Indian Head, MD 20640-5035"

Add: "DISTRIBUTION STATEMENT A: Approved for public release; distribution is unlimited."

* PAGE 2

2.2: Delete: "ASTM D 94-71 Saponification Number of Petroleum Products"

Add: "ASTM D 1045-86 Sampling and Testing Plasticizers Used in Plastics

ASTM D 1209-84 Color of Clear Liquids (Platinum-Cobalt Scale)"

* PAGE 4

4.5: Delete "in accordance with MIL-STD-1218" at end of paragraph.

4.5.1: Delete:

"a. Transfer 1.00 ± 0.01 gram(g) of iodine and 5.0 ± 0.1 g of potassium iodide to a 150-mL beaker. Add 25 mL of distilled water to dissolve the iodine and potassium iodide. Transfer the solution quantitatively to a 1000-mL volumetric flask, dilute to the mark with distilled water, and mix thoroughly.

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NOTE: Do not use solution after 24 hours.

b. Fill a clean 50-mL Nessler tube to the mark with sample plasticizer. Fill another 50-mL Nessler tube to the mark with the standard color solution (a. above). Hold the tubes side by side and compare their colors using transmitted light while looking downward through the tubes.

c. The sample shall be rejected if the color is darker than that of the standard solution."

Substitute:

"a. Prepare a No. 50 Platinum-Cobalt Color Standard in a 100-mL Nessler tube as described in ASTM D 1209-84 (10 mL No. 500 Platinum-Cobalt Stock Solution diluted to 100 mL). The No. 500 Platinum-Cobalt Stock Solution, already prepared, is available from laboratory supply houses. It may be used provided it meets the absorbance tolerance limits specified in ASTM D 1209 for the No. 500 Platinum-Cobalt Stock Solution.

b. Fill a clean 100-mL Nessler tube to the mark with the plasticizer sample.

c. Hold the sample and standard tubes side by side, and compare their colors using transmitted light while looking downward through the tubes.

d. The sample shall be rejected if the color is darker than that of the No. 50 Platinum-Cobalt Standard Solution."

* PAGE 5

4.5.5: Delete first sentence and substitute: "To 50 mL of absolute ethanol in a 125-mL flask, add 2 drops of 1-percent phenolphthalein indicator."

Delete "0.1 normal" and substitute "standardized 0.01 normal" throughout paragraph (2 places).

Add at end of paragraph: "If the titration exceeds 10 mL, repeat the determination using standardized 0.1 normal sodium hydroxide."

PAGE 6

* 4.5.6: Delete entire paragraph and substitute:

"4.5.6 Saponification number. The saponification number shall be determined in accordance with the titrimetric ester content procedure in ASTM D 1045-86 using a 3-hour reflux time except: 10 drops of 1 percent phenolphthalein indicator solution shall be used in lieu of bromothymol blue indicator; the 0.5 N KOH need not be standardized; 24-inch air condensers with ground glass joints may be used in place of water-cooled condensers; a 50-mL Class A burette with 0.1-mL graduations shall be used to deliver the HCl; and the glassware must be chemically clean. It is recommended that the normality of the HCl, standardized to four decimal

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places, be slightly higher than 0.50 N (but not greater than 0.52 N) so the burette will not require refilling during titration of the blank. The saponification number calculation requires subtraction of the acid number of the sample. To convert the percent acidity (determined in 4.5.5) to acid number, multiply by 9.35. The saponification number shall be reported to the nearest whole number. Duplicate results should be considered suspect if they differ by more than three saponification numbers."

4.5.8: After "Ubbelohde viscometer" add "Cannon-Fenske viscometer, or equivalent".

PAGE 7

Delete 6.5 and add:

"6.5 Supersession data. This specification supersedes MIL-P-23443A(WEP), dated 22 April 1965, with Amendment 1, dated 7 July 1977. This specification also includes the requirements of and supersedes Naval Sea Systems Command purchase description WS 2237, dated 11 October 1962."

* NOTE: The margins of this amendment are marked with asterisks to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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